Stephen Hawking

Born on January 8, 1942, English physicist Stephen Hawking became fascinated by space and theories about its nature and phenomena. Battling through adversity, he has made many brilliant contributions to astronomy and our understanding of the Universe.

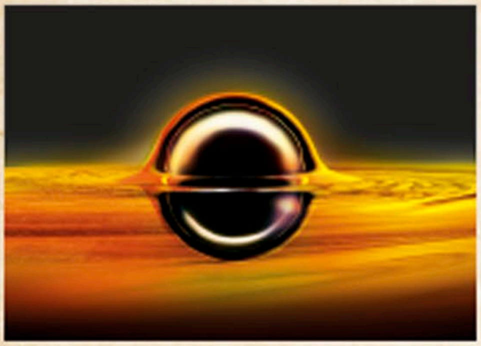
University life
Hawking (waving a handkerchief) with his fellow boat club members at the University of Oxford, in 1961. He studied physics and chemistry at Oxford before moving to Cambridge in 1962 to study cosmology—the study of the origins and development of the Universe.

A shocking diagnosis

In 1963, while studying at the University of Cambridge, Hawking was diagnosed with a disease called ALS (Amyotrophic Lateral Sclerosis), which destroys nerve cells. Initially given less than three years to live, he survived, although he was confined to a wheelchair from 1969 and lost his voice in 1985. Hawking communicates using a computer linked to a speech synthesizer that produces artificial speech.

Investigating black holes

Hawking began researching the incredibly dense remains of collapsed stars known as black holes. He suggested that they could be viewed as a smaller version of the Big Bang (the way many scientists believe the Universe began from a single point), but working in reverse.



Event horizon
Hawking's research on black holes enhanced the idea of a boundary (the edge of the "bubble" in the image above) around a black hole called an event horizon. Light or matter (yellow) crossing this boundary from the outside is pulled into the black hole by its incredibly strong gravity.

Hawking radiation

Scientists had thought that absolutely nothing could escape the immense gravitational pull of a black hole. However, in 1974, Hawking showed, in theory, how matter in the form of subatomic particles could be emitted from a black hole. This emission became known as Hawking radiation. This theory means that black holes do not exist forever, but gradually fade as they lose their energy.

A theory of everything

Hawking was the Lucasian Professor of Mathematics at Cambridge University from 1979 to 2009, a post once held by English physicist Isaac Newton. His later research sought a single, unifying theory to explain how the Universe works at both its biggest and smallest levels.

> **"My goal is simple. It is a complete understanding of the Universe, why it is as it is and why it exists at all."**
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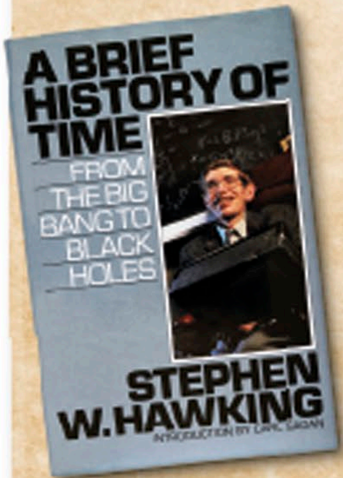
> **Stephen Hawking, quoted in the book *Stephen Hawking's Universe*, 1985**



Bestselling author
Hawking has authored hundreds of papers and more than a dozen books, including his 1988 bestseller, A Brief History of Time. This popular science guide to the Universe sold more than 10 million copies and was translated into 40 languages.

Zero gravity
In 2007, Hawking experienced zero gravity (weightlessness in space) on board a modified Boeing 727 aircraft that dives and climbs steeply to create a short period of weightlessness. "The zero-G part was wonderful… I could have gone on and on," he exclaimed afterward.